



B2 Proteins

B · Form and function · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Proteins are polymers of amino acids linked by peptide bonds formed through condensation reactions. Primary structure is the amino acid sequence; secondary structure (α -helices, β -pleated sheets) arises from hydrogen bonding along the backbone; tertiary structure is the overall 3D folding of a polypeptide stabilised by hydrogen, ionic and disulfide bonds and hydrophobic interactions; quaternary structure involves two or more polypeptide chains associating. Because sequence determines shape, and shape determines function, denaturing agents (extreme heat or pH) that disrupt these bonds change protein shape and destroy function, such as enzyme activity.

Worked example: follow the reasoning

Predicting the effect of lowering pH on an enzyme's activity. Step 1: at low (acidic) pH, excess H^+ ions interact with charged side chains (e.g. carboxylate and amine groups) that normally form ionic bonds stabilising tertiary structure. Step 2: this disrupts the ionic bonds, altering the folded shape of the polypeptide, including the active site. Step 3: substrate can no longer bind the active site with a complementary shape. Conclusion: below its optimum pH, the enzyme's rate of reaction decreases and eventually reaches zero as it becomes denatured.

Quick check · multiple choice

What is the main cause of enzyme denaturation by heat or extreme pH?

- A Breaking of peptide bonds in the primary structure
- B Breaking of hydrogen, ionic bonds and other interactions maintaining secondary/tertiary structure
- C Hydrolysis of the enzyme into free amino acids
- D An increase in the number of amino acids in the chain

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 An enzyme's rate of reaction was measured at temperatures from 10°C to 60°C . [7]
Rate increased steadily up to 40°C , then fell sharply between 40°C and 60°C .
(a) Explain the increase in rate between 10°C and 40°C in terms of kinetic energy and molecular collisions. [2]
(b) Explain the sharp decrease in rate above 40°C in terms of protein structure. [3]
(c) Predict and explain how the graph would differ for an enzyme from a bacterium living in a hot spring at 70°C . [2]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) As temperature increases, both enzyme and substrate molecules gain kinetic energy and move faster, increasing the frequency of successful collisions between substrate and active site, so the rate of reaction increases.

(b) Above the optimum temperature, the increased kinetic energy causes vibration that breaks the hydrogen and ionic bonds maintaining the enzyme's tertiary structure; this changes the shape of the active site so substrate can no longer bind effectively (denaturation), causing a sharp fall in rate.

(c) The hot-spring enzyme would likely show a similar-shaped curve but shifted to higher temperatures, with its optimum around 70°C and denaturation occurring only above this, because its tertiary structure is stabilised by additional/stronger bonds (e.g. more ionic or disulfide bonds) adapted to remain stable at higher temperatures.

Quick check answer

B. Denaturation disrupts the non-peptide bonds and interactions (hydrogen, ionic, hydrophobic) that hold the folded 3D shape, changing the active site shape, without breaking the peptide-bonded primary sequence.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.